

Random Forest in ML

~~Base~~ Random Forest is a type in Machine learning which is nothing but the combination of multiple decision trees working together.

It is used for both ~~sup~~ classification and regression.

It is a type of supervised machine learning.

Eg: Asking 100 doctors instead of just 1
Each doctor gives their opinion.
Final answer = Majority vote wins.

Advantage

- 1) High Accuracy: Better than single tree
- 2) No overfitting: Many trees fix errors
- 3) Find important Features: knows what matter

How it works

- 1) Take Original Data
- 2) Create random samples
- 3) Build tree for each sample
- 4) Each tree votes
- 5) Majority wins

Q: Dataset

ID	Age	Income	Student?	Buy?
1	Young	Low	No	No
2	Young	"	Yes	"
3	Middle	"	No	Yes
4	Senior	Medium	No	"
5	"	High	Yes	"
6	"	Medium	Yes	"
7	Middle	High	Yes	"
8	Young	Medium	No	No
9	"	High	Yes	Yes
10	Senior	Medium	Yes	Yes

Different trees from dataset

Tree	Looks at	Vote
Tree 1	Age, Income	No
Tree 2	"	"
Tree 3	Income, Student	"
Tree 4	Age, Income, Student	"
Tree 5	Age, Income	Yes

Vote count $\rightarrow$ No = 4

Yes = 1

Final Decision = No (since majority of vote for No)